HIBA V S

Machine Learning Engineer | Data Scientist

+91-6235899239 hibasareef@gmail.com portfolio LinkedIn GitHub Thrissur, Kerala, India

Summary

Final-year B.Tech Computer Science (Data Science) student with research experience in deep learning and computer vision at IIIT Kottayam, plus end-to-end ML projects spanning data analysis, model development, and deployment. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, and Pandas, with experience deploying models via FastAPI and React. Seeking ML Engineer / Data Scientist roles building scalable, real-world AI systems.

Research Experience

Research Intern — IIIT Kottayam **May 2026 – Present**

Satellite Disaster Assessment: MS-DCAFNet++ | PyTorch, xBD, Remote Sensing *Thrissur, India*

- Designed **MS-DCAFNet++**, a novel Siamese encoder-decoder for pixel-wise building damage segmentation across five severity classes on bi-temporal satellite image pairs.
- Introduced **DCAF (Damage-Conditioned Adaptive Fusion)**: an asymmetric attention module where the post-disaster image queries pre-disaster context, addressing a key limitation of symmetric models (CBAM, SE-Net).
- Replaced max-pooling with **Haar wavelet downsampling** and engineered a **hybrid Focal-Dice-Contrastive Boundary loss** to handle severe class imbalance (<1% destroyed pixels).
- Achieved a preliminary **mIoU of 0.37** and **F1-Destroyed of 0.56** on xBD (22,000+ image pairs); manuscript submitted to *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*.

Projects

FaceGuard: Real-Time Deepfake Detection System | *React, MobileNetV2, TensorFlow, FastAPI* **July 2025 – Feb 2026**

- Engineered a **14 MB MobileNetV2** edge-deployment model (vs. an 88 MB XceptionNet baseline) achieving **0.04 s per-frame CPU inference**, eliminating recurring cloud GPU costs for on-premise SME deployment.
- Achieved **83.6% test accuracy** on FaceForensics++ via transfer learning on ImageNet weights.
- Built a modular preprocessing pipeline (MediaPipe ROI crop → 224×224 resize) with a **temporal sliding-window buffer** to suppress false positives under motion blur.
- Delivered a full-stack React dashboard (static image, SSE video streaming, live WebCam via WebSocket) backed by FastAPI REST APIs, deployed on Vercel/Render.

Global Air Quality Intelligence System | *Python, Scikit-learn, Pandas, Matplotlib* **2026**

- Built an end-to-end ML pipeline analyzing AQI trends across **15 major cities (2015–2023)**; performed EDA on PM2.5, NO_2, SO_2, O_3 and weather features.
- Engineered domain-specific features (**pollutant load index, wind dispersion proxy**) to improve predictive signal beyond raw pollutant readings.
- Trained a **Random Forest classifier (86% accuracy, 6-class AQI category)** and a **Gradient Boosting regressor (R^2 = 0.94, MAE ≈ 8 AQI units)**.

SpaceX Falcon 9 Landing Prediction (IBM Capstone) | *Python, SQL, Folium, Plotly Dash, Scikit-learn* **2026**

- Collected launch data via the **SpaceX REST API** and **web scraping** (BeautifulSoup); cleaned and explored it with Pandas and SQL to identify drivers of landing success (payload, orbit, launch site, booster version).
- Built an interactive **Folium** map of launch-site proximities and a **Plotly Dash** dashboard for success rate by site and payload range.
- Tuned Logistic Regression, SVM, Decision Tree, and KNN with **GridSearchCV**, comparing models on held-out test accuracy.

Technical Skills

Programming Languages: Python, Java, SQL
Machine Learning & AI: PyTorch, TensorFlow/Keras, Scikit-learn, Random Forest, Gradient Boosting, OpenCV, MediaPipe, Computer Vision, Deep Learning, Feature Engineering
Data Science: Pandas, NumPy, SciPy, Matplotlib, Plotly Dash, Folium, BeautifulSoup, Exploratory Data Analysis (EDA), Data Visualization
Web Development: React.js, Next.js, FastAPI, REST APIs, WebSocket, HTML5, CSS3, Tailwind CSS
Tools & Platforms: Git, Docker, PostgreSQL, Vercel, Render

Certifications

IBM Data Science Professional Certificate (2026) | Generative AI Model Building — NxtWave (2025) | Learning Java — Infosys Springboard (2024)

Education

B.Tech Computer Science & Engineering (Data Science) **2023 – 2027**

APJ Abdul Kalam Technological University, IES College of Engineering *CGPA: 8.28 / 10*